MA 403 Real Analysis
Homework 5

1. Consider the two metrics on $\mathbb{R}^n$ defined in HW 4:

$$d_1(x,y) = \max_{1\leqslant i\leqslant n} |x-y|, \qquad d_2(x,y) = \sum_{i=1}^{n} |x_i - y_i|.$$

Prove that d_1 and d_2 satisfy the following inequalities for all $x, y \in \mathbb{R}^n$:

$$d_1(x,y) \leqslant \|x-y\| \leqslant d_2(x,y) \qquad \text{and} \qquad d_2(x,y) \leqslant \sqrt{n}\,\|x-y\| \leqslant n\, d_1(x,y).$$

2. Let (M,d) be a metric space, and let S, T be subsets of M such that $S \subset T$. Prove that

(a) $\overline{S} \subset \overline{T}$. (b) $\operatorname{int}(S) \subset \operatorname{int}(T)$.

3. Let (M,d) be a metric space, and let A, B, C be subsets of M such that A is dense in B and B is dense in C. Prove that A is dense in C.

4. Let A and B denote arbitrary subsets of a metric space M.

(a) Give an example in which $\operatorname{int}(\partial A) = M$.

(b) Give an example in which $\operatorname{int}A = \operatorname{int}B = \emptyset$ but $\operatorname{int}(A \cup B) = M$.

5. Using the **definition of the limit**, prove that:

(a) $\frac{x^n}{n!} \to 0$ for all $x \in \mathbb{R}$.

(b) If $\{x_n\}$ is a sequence such that $x_n \geqslant 0$ for all $n \in \mathbb{Z}_+$ and $x_n \to a$, then $\sqrt{x_n} \to \sqrt{a}$.

6. In a metric space (S,d), suppose that $x_n \to x$ and $y_n \to y$. Prove that $d(x_n, y_n) \to d(x,y)$.

7. Let $f : [a,b] \to \mathbb{R}$ be a continuous function such that $f(x) = 0$ when x is rational. Prove that $f(x) = 0$ for every $x \in [a,b]$.

8. Let f, g be defined on $[0,1]$ as follows:

$$f(x,y) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ 1 & \text{if } x \text{ is rational.} \end{cases} \qquad g(x,y) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ x & \text{is } x \text{ is rational.} \end{cases}$$

Prove that f is not continuous anywhere in $[0,1]$, and that g is continuous only at $x = 0$.